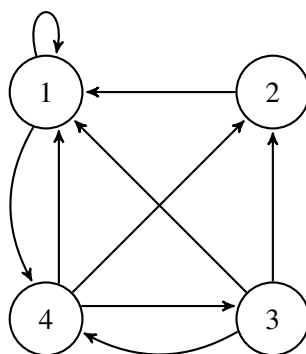


1. Adjacency and Connectivity Matrices

Consider the directed graph below. An arrow from node j to node i means that page j links to page i .



Graph B

- (a) Find the adjacency matrix A_B for graph B. Recall that for a directed graph,

$$(A_B)_{ij} = \begin{cases} 1 & \text{if there is an edge from node } j \text{ to node } i, \\ 0 & \text{otherwise.} \end{cases}$$

- (b) Each page splits one vote equally among the pages it links to, including itself if it has a self-loop. The connectivity matrix entry $(C_B)_{ij}$ is the share of page j 's vote that goes to page i . Find C_B by normalizing the columns of A_B . Check that each column sums to 1, and explain why this follows from the voting rule. A nonnegative matrix whose columns each sum to 1 is called *column stochastic*.

2. Mechanical Least Squares

We observe the following data and would like to fit a line:

$$\begin{array}{c|cccc} a_i & 2 & 4 & 6 & 8 \\ \hline b_i & 2 & 6 & 7 & 8 \end{array} \quad \mathbf{a} = \begin{bmatrix} 2 \\ 4 \\ 6 \\ 8 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 2 \\ 6 \\ 7 \\ 8 \end{bmatrix}.$$

First consider the model $b_i \approx sa_i$, which forces the fitted line to pass through the origin. The slope s is the only unknown parameter.

- (a) Write the squared error $L(s) = \sum_{i=1}^4 (b_i - sa_i)^2$ as a quadratic in s . Differentiate to find the best slope $\hat{s}$, and justify that it gives a minimum. Compute the minimum squared error.

- (b) Express $L'(s)$ using $\mathbf{a}$ and the residual $\boldsymbol{\varepsilon}(s) = \mathbf{b} - s\mathbf{a}$. What does $L'(\hat{s}) = 0$ say geometrically? Compute $\boldsymbol{\varepsilon}(\hat{s})$ and verify this condition directly.

- (c) Now fit the model $b_i \approx sa_i + c$, allowing an intercept c . Write the parameter vector as

$$\mathbf{x} = \begin{bmatrix} s \\ c \end{bmatrix}.$$

Construct the model matrix A so that the matrix-vector product $A\mathbf{x}$ contains all four predictions. Use the normal equations to find $\hat{s}$ and $\hat{c}$. Compute the squared error and compare it with part (a).